## 3 Numerical methods and comparison protocol

Section 2 showed that $A$ is sparse, symmetric and positive definite, with sensitivity measured by $\kappa_2(A)$. These properties decide which solvers apply and how fast they converge. Let $N = |F|$ be the number of unknowns, and $x_k$, $e_k = x - x_k$ and $r_k = b - Ax_k$ the $k$-th iterate, error and residual.

A direct method factorizes $A$ as $PA = LU$ or, for symmetric positive definite $A$, as $A = LL^T$ (Cholesky) at half the cost, then solves two triangular systems [5]; forming $A^{-1}$ costs more and is less accurate. Iterative methods avoid the extra nonzeros (fill-in) of sparse factors and only multiply by $A$.

A stationary iteration corrects $x_k$ with the residual, using an easily inverted approximation $M$ of $A$: $x_{k+1} = x_k + M^{-1}r_k$, so $e_{k+1} = (I - M^{-1}A)\,e_k$. It therefore converges for every $x_0$ if and only if the spectral radius $\rho(I - M^{-1}A)$ is below one [3]. Jacobi uses $M = D_A$, the diagonal of $A$. Its iteration matrix is similar to $I - S$ with $S = D_A^{-1/2} A D_A^{-1/2}$, so

$$\rho_J = \max_i |1 - \lambda_i(S)|. \tag{10}$$

As $\lambda_i(S) > 0$, Jacobi converges if and only if $\lambda_{\max}(S) < 2$. Damped Jacobi, $M = D_A/\omega$, converges if and only if $0 < \omega < 2/\lambda_{\max}(S)$; we use $\omega = 0.5$. Gauss–Seidel takes $M$ as the lower triangle of $A$ and converges for every symmetric positive definite $A$ [3].

The remaining methods exploit that $x$ minimizes $\phi(y) = \frac{1}{2}y^T A y - b^T y$, the elastic energy of Eq. (8) minus the work of the loads. Steepest descent minimizes $\phi$ along $r_k = -\nabla\phi(x_k)$; each step multiplies the energy norm of the error, $\|e\|_A = (e^T A e)^{1/2}$, by at most $(\kappa - 1)/(\kappa + 1)$, where $\kappa = \kappa_2(A)$. The conjugate gradient (CG) method instead minimizes $\|e_k\|_A$ over $x_0 + \mathcal{K}_k$, where $\mathcal{K}_k = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1}r_0\}$ is the Krylov space. It needs one product by $A$ per step and satisfies [4]

$$\|e_k\|_A \le 2\left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^k \|e_0\|_A, \tag{11}$$

so its iteration count grows like $\sqrt{\kappa}$ rather than $\kappa$. Diagonally preconditioned CG (PCG) is equivalent to CG applied to $S$, so $\kappa_2(S)$ replaces $\kappa$; it helps only if the diagonal of $A$ varies strongly. MINRES minimizes $\|r_k\|_2$ over the same space [1]. Full GMRES does so for any $A$ but stores all $k$ basis vectors [2], so it shows the cost of ignoring symmetry.

**Comparison protocol.** Every method receives the same $A$ and $b$, starts from $x_0 = 0$ and is accepted only if its recomputed relative residual satisfies

$$\eta(\hat{x}) = \frac{\|b - A\hat{x}\|_2}{\|b\|_2} < 10^{-8}. \tag{12}$$

Reaching the cap of 20,000 iterations, or $\eta > 10^{10}$ for ordinary Jacobi, counts as failure. Accuracy is the relative error $\|\hat{x} - x_{\mathrm{ref}}\|_2 / \|x_{\mathrm{ref}}\|_2$ against the sparse LU solution $x_{\mathrm{ref}}$ of the same system. Times are the best of three runs, excluding assembly and diagnostics.

Since $\hat{x}$ solves exactly the system with load perturbed by its residual, the standard bound [5]

$$\frac{\|\delta x\|_2}{\|x\|_2} \le \kappa_2(A)\,\frac{\|\delta b\|_2}{\|b\|_2} \tag{13}$$

limits its relative error to $\kappa_2(A)\,\eta$. This *algebraic* error measures how well one mesh's equations are solved. *Discretization* error, between finite element and continuum displacements, does not shrink with the tolerance; we estimate it by comparing meshes at matched physical points.

We also check that supports hold, that loads and the reactions of Eq. (7) balance in force and moment, and a *patch test*: a plate on rollers pulled by a uniform traction $p$ has the exact linear solution $u_x = px/E$, $u_y = -\nu p y/E$, which linear triangles must reproduce to rounding on any mesh.

## References


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